A machine-checked spectral floor for Pauli contextuality, and new configurations that attain it

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Priority note. The inequality formalized here was first posted by B. W. Jonah on 21 October 2026 (Zenodo, doi:10.5281/zenodo.23538829), where it is called the *spectral lower bound* or *spectral floor*. We derived it independently on 9 November 2026 and learned of the earlier notes the same day, during our own literature audit. Priority for the bound, for its saturation by Hoggar-type states in the three classical families, and for the exact four- and five-qubit values of those families belongs to the works cited in Section 1.2. What is new here is stated in Section 1.1; the results of Section 5 were found after that audit and were checked against the same sources.

Abstract

A noncontextual hidden-variable model assigns a value ± 1 to every N-qubit Pauli observable. The *contextuality degree* of a set of observables is the least number of its contexts (commuting triples with product $\pm I$) that such an assignment must get wrong. We give a kernel-checked Lean 4 proof, for every N and for arbitrary sets S, of the spectral floor first posted by B. W. Jonah: the number of satisfied minus violated contexts is at most $(D-2)\,|S|^{3/2}/(6\sqrt{D-1})$, where $D = 2^N$. The floor is attained exactly when a pure state has Pauli expectation values of one modulus on S and zero elsewhere. We exhibit such configurations outside the classical families and determine their degrees exactly: 135 four-qubit observables with 1095 contexts (degree 75) and 496 five-qubit observables with 10880 contexts (degree 480), both proved in Lean with kernel-evaluated counts, and 8128 seven-qubit observables with 2 824 192 contexts (degree 729 344). The Pauli support of any such state is a two-character set, hence gives a strongly regular graph, and a bent function in the three-, five- and seven-qubit cases. By exhaustive enumeration we classify the states $U|+\cdots+\rangle$, with U a diagonal gate of the third level of the Clifford hierarchy, that have this property on up to five qubits: there is none on four qubits and one class on five, a circuit of three controlled-S and two CCZ gates whose configuration is Clifford-equivalent to the support of a state found numerically by Cieśliński. On seven qubits a circuit of three controlled-S and eight CCZ gates gives the configuration above. For the classical families the checked corollaries include a violated fraction of at least $1/2 - \sqrt{D+1}/(D+2)$ for the full N-qubit configuration, for every N. The work was produced with an AI research system under the author's direction, as disclosed in the paper.

1. Scope

1.1 What is new in this work

1. **A formal proof of the floor.** The spectral floor is proved in Lean 4 for every number of qubits $N \geq 1$ and for *arbitrary* sets of observables with their induced contexts (theorem

`hoggar_bound`). The proof is about the actual matrices: Pauli operators are defined as explicit complex matrices indexed by bit vectors, their product phases and commutation rule are theorems, and the bound is derived from the spectral theorem. Earlier accounts of the bound are explicitly not formally verified [J2], and the earlier Lean development of Li, Gupta and Ganesh [LGG] covers one four-qubit configuration by a different (combinatorial) argument.

2. **Exact degrees outside the classical families** (Section 5). The published statements of the floor and of its equality case concern families closed under products of commuting pairs: the full space and the quadrics [J2]. The trace argument needs no closure, and the floor is attained on sets that are not closed. We determine two such degrees exactly: 75 for a 135-point four-qubit configuration with 1095 contexts, and 480 for a 496-point five-qubit configuration with 10880 contexts. Both are proved in Lean in full (theorems `pivot_degree` and `cieslinski5_degree`): the lower bound is an instance of `hoggar_bound`, the upper bound an explicit assignment, and the counts are kernel evaluations.
3. **A four-qubit state.** The state ψ_B of the abstract has Pauli expectation values $\pm 1/3$ on 135 observables and 0 on the other 120, like the known state $(-3, 1, \ldots, 1)/\sqrt{24}$, but its support is not a quadric. We classify all 4^6 states with fourth-root-of-unity amplitudes on the same six basis states: they fall into exactly two classes, according to the parity of the number of imaginary amplitudes (Proposition 6).
4. **A saturation criterion for arbitrary sets** (Theorem 4), in spectral, state and purely combinatorial form; the observation that for every such state the violated contexts are the pairs of largest quantum metric (Proposition 8); and the fact that the support of such a state is a two-character set, so that it defines a strongly regular graph, and a bent function when the modulus is $2^{-(N-1)/2}$ (Proposition 10).
5. **Flat phase states** (Section 5.3). A state $U|+\cdots+\rangle$ with U a product of T, controlled-S (CS) and CCZ gates has a single non-zero modulus exactly when a symmetric trilinear form over $\mathbb{F}_2$ has constant rank (Proposition 11). We enumerate all such forms on up to five qubits (Theorem 12): none on four qubits, one class on five. The five-qubit class gives a five-gate circuit for a state with the Pauli support of [Ci]'s numerically found s_5, up to a Clifford operation.
6. **Seven qubits** (Theorem 13). A circuit of three CS and eight CCZ gates prepares a state with expectation values $\pm 1/8$ on 8128 observables and 0 on the others. The configuration has 2 824 192 contexts and contextuality degree exactly 729 344. Within the family of forms we could enumerate completely (about 1.5×10^{10} forms) it is unique up to CNOT circuits.
7. **Formally verified numerical corollaries for the classical families**, listed in the abstract and in Table 2, including formal answers to two questions asked in [LGG]: the degree of $Q^+(9, 2)$ exceeds 4743, and the violated fraction of the four-qubit full space and of the five-qubit hyperbolic quadric is strictly above 1/5. The exact values 63 (three qubits) and 315 (four-qubit hyperbolic quadric) are also proved in full, by the spectral route.
8. **Consequences spelled out for all** N: an exponentially growing quantum-to-classical ratio with a proof (the 2010 argument for such growth rests on a lemma that fails at three qubits), a noise threshold that tends to zero, and a two-sided estimate of the optimal violated fraction (Proposition 4).
9. **An independent reproduction and a running demonstration**: a search engine, four independent implementations of the geometry (C, Python, JavaScript, Eshkol) and the Lean model agree on every count; the experiments run on the Tsotchke stack with each component labelled for what it is. The five- and seven-qubit states are prepared there as circuits.

1.2 What is not new, and whose it is

- The contextuality degree and its first values: de Boutray, Holweck, Giorgetti, Masson and Saniga [dB]; Muller, Saniga, Giorgetti, de Boutray and Holweck [M1] (three-qubit degree 63, the split Cayley hexagon of violated contexts); Muller, Saniga, Giorgetti, Holweck and Kelleher [M2] (upper bounds for four to seven qubits, lower bound $L/5$); Saniga et al. [S] (hexagons).
- The exact value 315 for the four-qubit hyperbolic quadric, its 240 optimal sets and a Lean proof of that value: Li, Gupta and Ganesh [LGG].
- The exact values $d(Q^-(7,2)) = 315$, $d(W(7,2)) = 1575$, $d(Q^+(9,2)) = 6975$ (computer-assisted), the operator $A = \sum_v f(v)P_v$ and its Hoggar spectrum $\{21, -3^7\}$: Jonah [J1].
- The spectral floor itself, its Kramers (quaternionic) form for elliptic quadrics, the description of quadrics by the antiunitaries P_cK, the operator identity for A^2 on sets closed under products, and the theorem that among the three families the floor is attained exactly for $W(5,2)$, $Q^+(7,2)$, $Q^-(5,2)$ (and trivially $Q^-(3,2)$): Jonah [J2], with the trace formula also in [J3]. Stronger five-qubit lower bounds ($d(W(9,2)) \geq 28830$, $d(Q^-(9,2)) \geq 6510$) are announced, with proofs held privately, in [J4]. Sets that are not closed under products are named in [J2] as not covered.
- Pure states whose Pauli expectation values are zero or of one modulus: the two-qubit state of Liu, Low and Yin [LLY]; the four-qubit state $(-3, 1, \ldots, 1)/\sqrt{24}$ and the five-qubit state $|s_5\rangle$ with 135 and 496 non-zero values, found numerically by Cieśliński [Ci], who calls a (possibly mixed) state *Pauli flat* when all of its Pauli expectation values have one modulus and lists these pure states under the relaxed condition; real four-qubit states of this kind as the optima of $Q^+(7,2)$ [J4]. Kang [K] proves that no four-qubit state, and no state of an odd number $N > 1$ of qubits, has all squared Pauli expectation values in $\{0, 2^{-N}\}$. None of [LLY, Ci, K] discusses contextuality, phase states, or more than five qubits.
- Diagonal gates of the Clifford hierarchy and their phase polynomials: Cui, Gottesman and Krishna [CGK]. $\mathbb{Z}_4$-valued quadratic forms and their character sums: [CCKS]. Two-character sets, two-weight codes and strongly regular graphs: Calderbank and Kantor [CK]. The bent functions of Maiorana and McFarland: [McF].
- The scalar inequality behind the floor is classical: it is Okumura's lemma [O] (equivalently the maximal sample skewness, Wilkins [W]).
- The Hoggar lines and the cubic purity condition for quantum states: Hoggar [H], Stacey [St], Jones and Linden [JL]. Bent functions and their difference sets: Rothaus [R]. Point sets with the hyperplane intersection numbers of a quadric (quasi-quadrics): De Clerck, Hamilton, O'Keefe and Penttila [DHOP].

Our derivation in Sections 3, 4 and 6 was made without knowledge of [J1]–[J4]; where it coincides with them we say so. We did not re-run the exhaustive computations behind the exact values of [J1] and do not vouch for them; we only observe that every assignment we found is consistent with them. A companion note in the same repository, written by a separate set of AI agent sessions, settles an instance of a conjecture of [J3] that is unrelated to the results here; it is not part of this paper.

2. Setting

Pauli operators. Let $D = 2^N$. A *point* is a nonzero vector $v = (a \mid c) \in \mathbb{F}_2^{2N}$. It labels the Hermitian operator

$$P_v \;=\; i^{\,|a \wedge c|}\, X^a Z^c,$$

a tensor product of I, X, Y, Z (the exponent $|a \wedge c|$ counts the factors equal to $Y = iXZ$). $P_0 = I$, $P_v^2 = I$, and $\operatorname{tr} P_v = 0$ for $v \neq 0$. Products close up to a phase,

$$P_u P_v = \tau(u,v)\, P_{u+v}, \qquad \tau(u,v) = i^{\,k_u + k_v + 3k_{u+v}}\, (-1)^{c_u \cdot a_v}, \quad k_v = |a_v \wedge c_v|,$$

and with the symplectic form $\langle u, v\rangle = c_u \cdot a_v + c_v \cdot a_u$ one has $\tau(v,u) = (-1)^{\langle u,v\rangle}\tau(u,v)$ and $\tau(u,v)^2 = (-1)^{\langle u,v\rangle}$. So P_u and P_v commute exactly when $\langle u, v\rangle = 0$, and then $\tau(u,v) = s(u,v) \in \{+1, -1\}$.

Contexts. A *context* (a line of the symplectic polar space $W(2N-1, 2)$) is a set $\ell = \{u, v, u+v\}$ of three distinct points with $\langle u, v\rangle = 0$. Its three operators commute and $P_u P_v P_{u+v} = s_\ell\, I$ with $s_\ell = s(u,v)$, which does not depend on the order.

Configurations and assignments. Fix a set S of points and let $\mathcal{L}(S)$ be the set of all contexts contained in S, $L_S = |\mathcal{L}(S)|$. S is *closed* if the product of any two commuting observables of S is again in S. An assignment $f : S \to \{+1, -1\}$ *violates* ℓ when $f(u)f(v)f(u+v) \neq s_\ell$. Put

$$\beta_S(f) = \sum_{\ell \in \mathcal{L}(S)} s_\ell\, f(u)f(v)f(u+v) = L_S - 2\,\mathrm{viol}(f), \qquad d(S) = \min_f \mathrm{viol}(f).$$

$d(S)$ is the contextuality degree [dB]. For the noncontextuality inequality $\chi = \sum_\ell s_\ell \langle P_u P_v P_{u+v}\rangle \leq L_S - 2d(S)$ quantum mechanics gives $\chi = L_S$ in every state [C, LGG].

Three families. The *full space* is $S = \mathbb{F}_2^{2N} \setminus \{0\}$, with $D^2 - 1$ points and $L = (D^2-1)(D^2-4)/12$ contexts. For $c \in \mathbb{F}_2^{2N}$ let $q_0(v) = |a \wedge c| \bmod 2$ and $Q_c = \{v \neq 0 : q_0(v) + \langle c, v\rangle = 0\}$. Q_c is a *hyperbolic* quadric $Q^+(2N-1, 2)$ if $q_0(c) = 0$ and an *elliptic* quadric $Q^-(2N-1, 2)$ if $q_0(c) = 1$. With K complex conjugation in the computational basis, $\theta_c = P_c K$ is antiunitary, $\theta_c P_v \theta_c^{-1} = (-1)^{q_0(v) + \langle c, v\rangle} P_v$ and $\theta_c^2 = (-1)^{q_0(c)}$. So the quadrics are the sets of Pauli observables fixed by a Pauli antiunitary: a real structure ($\theta^2 = 1$) for hyperbolic quadrics and a quaternionic structure ($\theta^2 = -1$) for elliptic ones [J2]. Q_0 is the set of real symmetric Pauli matrices (an even number of Y factors). All three families are closed.

3. The spectral floor

Theorem 1 (spectral floor; Jonah [J2] for the three families; formalized here for arbitrary S). Let $N \geq 1$, let S be any set of points and $f : S \to \{\pm 1\}$. Then

$$\beta_S(f) \;\leq\; \frac{(D-2)\,|S|^{3/2}}{6\sqrt{D-1}}, \qquad \text{hence} \qquad d(S) \;\geq\; \frac{L_S}{2} - \frac{(D-2)\,|S|^{3/2}}{12\sqrt{D-1}}.$$

Proof. Let $A = \sum_{v \in S} f(v) P_v$, a Hermitian $D \times D$ matrix.

(i) Three traces. $\operatorname{tr} A = 0$ because $0 \notin S$. Since $\operatorname{tr}(P_u P_v) = D\,\delta_{uv}$ and $f(v)^2 = 1$, $\operatorname{tr} A^2 = D\,|S|$. Since $\operatorname{tr}(P_u P_v P_w) = D\,\tau(u,v)\,[w = u+v]$,

$$\operatorname{tr} A^3 = D \sum_{\substack{u,v \in S \\ u+v \in S}} f(u)f(v)f(u+v)\,\tau(u,v).$$

The pairs with $\langle u, v\rangle = 1$ cancel against their transposes, because $\tau(v,u) = -\tau(u,v)$. The pairs with $\langle u, v\rangle = 0$ are the ordered pairs of distinct points of a context ($u = v$ would give $u + v = 0 \notin S$), six for each context, each contributing $s_\ell f(u)f(v)f(u+v)$. Therefore $\operatorname{tr} A^3 = 6D\,\beta_S(f)$.

(ii) A scalar inequality (Okumura [O]). Let $x_1, \ldots, x_n$ be real with $\sum x_i = 0$ and $\sum x_i^2 = nm$, $n \geq 2$, and let $a = \sqrt{(n-1)m}$. By the Cauchy–Schwarz inequality applied to the other $n - 1$

numbers, $x_i^2 \le (n-1)(nm - x_i^2)$, so $x_i \le a$ for every i. Hence $(x_i - a)\big((n-1)x_i + a\big)^2 \le 0$. Summing over i and using the two constraints,

$$(n-1)\sum_i x_i^3 \;\le\; n\,m\,(n-2)\,a.$$

(iii) Conclusion. Apply (ii) to the eigenvalues of A with $n = D$, $m = |S|$: $(D-1)\cdot 6D\beta \le D\,|S|\,(D-2)\sqrt{(D-1)|S|}$. ∎

Three remarks. First, no closure property of S is used: products that leave S do not contribute to the trace. Second, the hypothesis $f = \pm 1$ enters only through $\operatorname{tr} A^2 = D|S|$, so the argument bounds the cubic form $T(x) = \sum_\ell s_\ell x_u x_v x_w$ on the sphere $\sum x_v^2 = |S|$. Third, the bound applies to the induced configuration only, that is, with *all* contexts contained in S.

Theorem 2 (quaternionic form; Jonah [J2]). If every P_v, $v \in S$, commutes with an antiunitary θ with $\theta^2 = -1$ (for instance, S lies in an elliptic quadric), then

$$\beta_S(f) \;\le\; \frac{(D-4)\,|S|^{3/2}}{6\sqrt{2(D-2)}}.$$

Proof. A is a real combination of operators commuting with θ, so $\theta A = A\theta$. If $Ax = \lambda x$ then $A\theta x = \lambda\theta x$, and $\langle x, \theta x\rangle = 0$ because $\langle \theta x, \theta y\rangle = \overline{\langle x, y\rangle}$ gives $\langle x, \theta x\rangle = \overline{\langle \theta x, \theta^2 x\rangle} = -\langle x, \theta x\rangle$. Every eigenspace is therefore an orthogonal sum of planes $\{x, \theta x\}$ and every eigenvalue has even multiplicity (Kramers). Apply (ii) to the $D/2$ distinct pair values, with the same m. ∎

Theorem 2 is not part of the Lean development; it is checked numerically (Supplementary Section S2).

Closed forms. With the standard counts, Theorems 1 and 2 give for the three families the same expression in different arguments. Writing $\varphi(D) = \frac{1}{2} - \sqrt{D+1}/(D+2)$,

family	points	contexts L	violated fraction at least
full space $W(2N-1,2)$	D^2-1	$(D^2-1)(D^2-4)/12$	$\varphi(D)$
elliptic quadric $Q^-(2N-1,2)$	$(D+1)(D-2)/2$	$(D+1)(D-2)(D+2)(D-4)/48$	$\varphi(D)$
hyperbolic quadric $Q^+(2N-1,2)$	$(D-1)(D+2)/2$	$(D-1)(D+2)(D-2)(D+4)/48$	$\varphi(D/2)$

The three cases are the complex, quaternionic and real members of Dyson's threefold way; the real family at N qubits has the same floor as the complex family at $N-1$ qubits.

4. Saturation in the classical families

Theorem 3 (equality; Jonah [J2, Theorem 6] for the three families). Equality holds in Theorem 1 if and only if there is a unit vector ψ with

$$\langle\psi|P_v|\psi\rangle = f(v)\sqrt{\frac{D-1}{|S|}} \quad (v \in S), \qquad \langle\psi|P_v|\psi\rangle = 0 \quad (v \notin S,\ v \neq 0).$$

Proof. Equality in (ii) forces $x_i \in \{a, -a/(n-1)\}$ for every i, and the zero sum forces exactly one value a. So $A = (a-b)\,|\psi\rangle\langle\psi| + b\,I$ with $b = -a/(D-1)$; reading off Pauli coefficients gives the claim, and the converse is the same computation backwards. ∎

We call such a ψ *flat on* S, after [Ci]. For the full space the condition says that ψ is a fiducial vector of a SIC covariant under the N-qubit Pauli group. Such vectors exist only for $N = 1$ and $N = 3$ (Godsil and Roy [GR]; an elementary proof through the integrality of the floor is in [J2]): for $N = 3$ they are the Hoggar lines.

The three saturated cases. The Hoggar fiducial $\psi_H \propto (-1+2i, 1, 1, 1, 1, 1, 1, 1)$ has $\langle P_v\rangle = \pm 1/3$ for all 63 three-qubit Pauli observables. We verified the following by exact counting in four independent implementations (Supplementary Section S2); the values themselves are known [M1, dB, LGG, J2].

configuration	assignment	violated	floor	spectrum of A
$W(5,2)$: 63 points, 315 contexts	signs of $\langle\psi_H\|P_v\|\psi_H\rangle$	63	63	$\{21, (-3)^7\}$
$Q^-(5,2)$: 27 points, 45 contexts	the same signs, restricted	9	9	$\{9^2, (-3)^6\}$
$Q^+(7,2)$: 135 points, 1575 contexts	signs of the realified $\psi_H \in \mathbb{R}^{16}$	315	315	$\{45, (-3)^{15}\}$

In the third row, $\mathbb{C}^8$ is read as $\mathbb{R}^{16}$; the resulting real four-qubit state has expectation $\pm 1/3$ on all 135 real Pauli observables and 0 on the 120 imaginary ones (this description of the optimal assignments of $Q^+(7,2)$ as sign patterns of real flat states is announced in [J4]; the state $(-3, 1, \ldots, 1)/\sqrt{24}$ of [Ci] is of this kind). The violated contexts form, respectively, a classically embedded split Cayley hexagon (63 lines, three per point, incidence graph of diameter 6 and girth 12, the three lines through each point in one totally isotropic plane), nine pairwise disjoint lines, and a copy of the dual polar space $DW(5,2)$ (315 lines, seven per point, collinearity graph distance-regular with intersection array $\{14, 12, 8; 1, 3, 7\}$), in agreement with [M1, S, M2, LGG].

By Theorem 3, the optimal assignments of $W(5,2)$ are exactly the sign patterns of Pauli-covariant SIC fiducials in dimension 8. Our enumeration finds 120 optimal violated sets in the orbit of the Hoggar pattern under $Sp(6,2)$, and the context–point incidence matrix has rank 56 over $\mathbb{F}_2$, so each set comes from $2^7 = 128$ assignments: $120 \cdot 128 = 15360 = 240 \cdot 64$ assignments, matching the 240 Hoggar-type SICs of 64 vectors each [St]. For $Q^+(7,2)$ the orbit of our violated set under the reflections of $O^+(8,2)$ has 240 elements, the number proved in [LGG].

Proposition 3 (violated contexts in the quantum metric). Let ψ be a state, $p_v = \langle P_v\rangle$, and let $Q_{uv} = \frac{1}{4}\big(\langle P_u P_v\rangle - p_u p_v\big)$ be its quantum geometric tensor along the flows $e^{-itP_v/2}$. For a context $\{u, v, w\}$ with sign s, the Fubini–Study metric is $g_{uv} = \frac{1}{4}(s\,p_w - p_u p_v)$. If $f = \mathrm{sign}(p)$, the context is violated exactly when $4|g_{uv}| = |p_w| + |p_u p_v|$ and satisfied exactly when $4|g_{uv}| = \big||p_w| - |p_u p_v|\big|$.

Proof. $\langle P_u P_v\rangle = s\,p_w$; the context is violated when $s\,p_w$ and $p_u p_v$ have opposite signs. ∎

For ψ_H: $g_{vv} = 2/9$; $|g_{uv}| = 1/9$ on the 189 commuting pairs lying on the 63 violated contexts, $1/18$ on the 756 pairs lying on satisfied contexts, $1/36$ on the 1008 anticommuting pairs; the Berry curvature has modulus $1/6$ on anticommuting pairs and vanishes on commuting ones. The hexagon is the set of pairs of largest off-diagonal metric (computed with the QGTL kernels, Supplementary Section S1).

5. Saturation outside the classical families

5.1 The criterion

The statements of this section concern arbitrary sets S, which need not be closed. For an assignment f and a point $w \in S$ let the *excess* $e_w(f)$ be the number of satisfied minus the

number of violated contexts of S through w. For a point $w \notin S$, $w \neq 0$, let the *outside sum* be $b_w(f) = \sum s(u,v)f(u)f(v)$ over the unordered pairs of commuting points u, v of S with $u + v = w$.

Theorem 4 (saturation criterion). Let S be a set of points, $f : S \to \{\pm 1\}$ and $A = \sum_{v \in S} f(v) P_v$. Then

$$A^2 \;=\; |S|\, I \;+\; 2 \sum_{w \in S} e_w(f)\, f(w)\, P_w \;+\; 2 \sum_{w \notin S,\; w \neq 0} b_w(f)\, P_w,$$

and the following are equivalent:

- (a) f attains the floor: $\beta_S(f) = (D-2)|S|^{3/2}/(6\sqrt{D-1})$;
- (b) A has the simple eigenvalue $\sqrt{(D-1)|S|}$ and the eigenvalue $-\sqrt{|S|/(D-1)}$ with multiplicity $D-1$;
- (c) some pure state is flat on S with signs f (Theorem 3);
- (d) $e_w(f) = (D-2)\sqrt{|S|}\,/\,(2\sqrt{D-1})$ for every $w \in S$, and $b_w(f) = 0$ for every $w \notin S$.

When they hold, $d(S) = \frac{1}{2}L_S - (D-2)|S|^{3/2}/(12\sqrt{D-1})$ exactly, f is optimal, and, for $N \geq 2$, $(D-1)|S|$ is a perfect square.

Proof. In $A^2 = \sum_{u,v} f(u)f(v)P_uP_v$ the diagonal terms give $|S|\, I$ and the two orders of an anticommuting pair cancel. A commuting pair $u \neq v$ contributes $2s(u,v)f(u)f(v)P_{u+v}$. If $w = u + v \in S$, the pair is a context through w and $s(u,v)f(u)f(v) = \pm f(w)$ according to whether it is satisfied; summing over the contexts through w gives $2e_w(f)f(w)P_w$. If $w \notin S$ the pairs give $2b_w(f)P_w$. This is the identity; for closed S the last sum is empty and it is Theorem 1 of [J2]. (a) $\Leftrightarrow$ (b) is the equality case of the scalar inequality, and (b) $\Leftrightarrow$ (c) is Theorem 3. Condition (b) says $(A - a)(A + a/(D-1)) = 0$ with $a = \sqrt{(D-1)|S|}$, that is $A^2 = |S|\, I + \frac{a(D-2)}{D-1} A$ (a traceless matrix with these two eigenvalues has the larger one exactly once); comparing Pauli coefficients with the identity gives (d). The floor is an integer when attained, which for $D > 2$ forces $\sqrt{(D-1)|S|}$ to be rational. ∎

Condition (d) replaces the *violation-regularity* of [J2, Theorem 6], to which it reduces when every point lies on the same number of contexts. For $N \leq 5$ the number $D - 1$ is squarefree, so a set that attains the floor has $|S| = (D-1)k^2$ and excess $(D-2)k/2$: $|S| \in \{3, 12\}$ for two qubits, $\{7, 28, 63\}$ for three, $\{15, 60, 135, 240\}$ for four, $\{31, 124, 279, 496, 775\}$ for five.

Table 3. The configurations we know to attain the floor (besides the stabilizer case $|S| = D - 1$, $d = 0$, a maximal commuting set with the signs of a stabilizer state).

state	N	observables	contexts	degree	modulus	excess	closed	state from
$(i, i, i, 1)/2$	2	12	8	0	1/2	2	no	[LLY]
CCZ applied to $(1, \ldots, 1)/\sqrt{8}$	3	28	56	0	1/2	6	no	folklore
Hoggar fiducial	3	63	315	63	1/3	9	yes	[H]
$(-3, 1, \ldots, 1)/\sqrt{24}$	4	135	1575	315	1/3	21	yes	[J4, Ci]
ψ_B (pivot state)	4	135	1095	**75**	1/3	21	no	this work
s_5	5	496	10880	**480**	1/4	60	no	[Ci]
Φ_5 (3 CS, 2 CCZ)	5	496	10880	**480**	1/4	60	no	this work
Φ_7 (3 CS, 8 CCZ)	7	8128	2824192	**729344**	1/8	504	no	this work

The degrees in bold are new, as is the reading of the first two rows: the 12 two-qubit observables outside a maximal commuting set, and the 28 real three-qubit observables that are not diagonal,

are noncontextual configurations on which the floor is nevertheless exact. The configurations of s_5 and Φ_5 are equivalent under the Clifford group (Theorem 12). The first seven rows are proved in Lean (Section 8); all eight are reproduced on Moonlab (Supplementary Section S1).

5.2 A four-qubit and a five-qubit configuration

Theorem 5 (the pivot configuration). Let

$$\psi_B \;=\; \tfrac{1}{\sqrt{6}}\big(\,|0011\rangle + |0111\rangle + |1011\rangle + |1100\rangle + |1101\rangle + i\,|1110\rangle\,\big),$$

with qubit 0 written on the right, and let S_B be the set of observables with non-zero expectation value in ψ_B.

1. The expectation values are $\pm 1/3$ on S_B, and $|S_B| = 135$: the 15 diagonal observables $(0 \mid c)$, and for each $a \neq 0$ the eight observables $(a \mid c)$ with $a \cdot c = \epsilon(a)$, where $\epsilon(a) = 1$ for the five values $a \in E = \{0010, 0011, 0101, 1001, 1101\}$ and $\epsilon(a) = 0$ otherwise.
2. S_B contains 1095 contexts: 35 through each diagonal observable and 23 through each of the other 120. It is not closed and it is not a quadric, but every hyperplane $w^\perp$ meets it in 71 points if $w \in S_B$ and in 63 points otherwise, as for $Q^+(7,2)$: it is a quasi-quadric in the sense of [DHOP]. It differs from the quadric $a \cdot c = 0$ in 80 points, those with $a \in E$, and E consists of five pairwise non-orthogonal vectors (an ovoid) of the doily $W(3,2)$ defined by the form $x_0y_1 + x_1y_0 + x_2y_3 + x_3y_2$.
3. $d(S_B) = 75$. The assignment given by the signs of the expectation values violates the 15 lines of that doily, read among the diagonal observables, and 60 further contexts, each with exactly one diagonal observable. Every non-diagonal observable of S_B lies on exactly one violated context and every diagonal one on seven.

Proof. Items 1 and 2 and the count of 75 violated contexts are finite computations in $\mathbb{Z}[i]$ (Supplementary Section S2); item 1 also follows from Proposition 6. The bound $d(S_B) \geq 75$ is Theorem 1 with $|S| = 135$, $D = 16$, $L = 1095$: $\beta \leq 14 \cdot 135 \cdot 45/(6 \cdot 15) = 945$. In Lean this is `pivot_degree`. ∎

The noncontextuality inequality of this configuration has classical bound 945 and quantum value 1095 in every state. The quadric $Q^+(7,2)$ has the same number of observables and the same classical bound, with quantum value 1575.

Proposition 6 (flat states on a bent support). Let T be the set of the six $x \in \mathbb{F}_2^4$ with $x_0x_1 + x_2x_3 = 1$, the support of a bent function, and let $\phi : T \to \mathbb{C}$ with $|\phi| = 1$. The state $\psi_\phi = 6^{-1/2} \sum_{x \in T} \phi(x)|x\rangle$ is flat on its Pauli support if and only if, up to a global phase, all $\phi(x)$ are fourth roots of unity. In that case the support has 135 points and the modulus is $1/3$; it is a hyperbolic quadric (1575 contexts, degree 315) when the number of imaginary amplitudes is even, and it has 1095 contexts and degree 75 when that number is odd. The two classes are not related by any Clifford operation.

Proof. T is a difference set with parameters $(16, 6, 2)$ [R]: every non-trivial character sum over T is ± 2, and for every $a \neq 0$ exactly one unordered pair $x_a, x_a + a$ lies in T. Hence $\langle Z^c \rangle = \pm 1/3$ for $c \neq 0$, and for $a \neq 0$, with $z_a = \overline{\phi(x_a + a)}\,\phi(x_a)$,

$$\langle X^a Z^c \rangle = \tfrac{1}{6}(-1)^{c \cdot x_a}\big(z_a + (-1)^{c \cdot a}\,\overline{z_a}\big),$$

so the expectation value of $P_{(a|c)}$ is $\pm\frac{1}{3}\operatorname{Re} z_a$ when $a \cdot c = 0$ and $\pm\frac{1}{3}\operatorname{Im} z_a$ when $a \cdot c = 1$. A common modulus must be $1/3$, which holds exactly when every z_a is ± 1 or $\pm i$, that is, when all

ratios of amplitudes are fourth roots of unity. Then for each $a \neq 0$ exactly the eight observables with $a \cdot c = \epsilon(a)$ are in the support, where $\epsilon(a) = 0$ if z_a is real and 1 if it is imaginary. Write $\phi = i^k$ and $\kappa = k \bmod 2$; then $\epsilon(x + x') = \kappa_x + \kappa_{x'}$. A diagonal Clifford operation multiplies ϕ by $i^{q(x)}$ for a $\mathbb{Z}_4$-valued quadratic form q and changes κ by the restriction to T of a linear functional; these sixteen restrictions have weights 0, 2 and 4, and together with the all-ones vector (a global factor i) they span the even-weight vectors of $\mathbb{F}_2^T$. So κ of even weight is equivalent to $\kappa = 0$, a real state, whose support is the quadric $a \cdot c = 0$; and κ of odd weight is equivalent to a single imaginary amplitude, as in ψ_B. Symplectic maps preserve the number of contexts, and $1575 \neq 1095$. The two counts were computed for all 32 supports that occur, over all 4096 phase patterns (Supplementary Section S2). ■

Theorem 7 (five qubits). Let S_5 be the set of the 496 five-qubit observables with non-zero expectation value in the state s_5 of Cieśliński [Ci, eq. (18)], all equal to $\pm 1/4$. Then S_5 contains 10880 contexts and $d(S_5) = 480$. The optimal assignment, the signs of the expectation values, leaves 16 observables on no violated context (60 contexts through each), 240 on two (of 64) and 240 on four (of 68). Every hyperplane $w^\perp$ meets S_5 in 256 points if $w \in S_5$ and in 240 otherwise, as for the complement of $Q^+(9, 2)$; S_5 is not such a complement.

Proof. The counts are finite computations in $\mathbb{Z}[i]$. The lower bound is Theorem 1 with $|S| = 496$, $D = 32$: $\beta \leq 30{\cdot}496{\cdot}124/(6{\cdot}31) = 9920 = 10880 - 2{\cdot}480$. In Lean this is `cieslinski5_degree`. No state vanishes on every observable of a hyperbolic quadric: these are the observables commuting with an antiunitary θ with $\theta^2 = 1$, and the condition would give $\rho + \theta\rho\,\theta^{-1} = 2I/D$, of rank D, for a projector ρ of rank one. ■

Proposition 8 (the optimum in the quantum metric). Let ψ be flat on S with modulus μ, and let f be the signs of its expectation values. For $u \neq v$ in S the Fubini–Study metric of Proposition 3 has $|g_{uv}| = (\mu + \mu^2)/4$ if $u, v, u + v$ form a context of S violated by f, $|g_{uv}| = (\mu - \mu^2)/4$ if they form a satisfied context of S, and $|g_{uv}| = \mu^2/4$ otherwise (u, v anticommute, or their product is outside S). So the violated contexts of the optimal assignment are exactly the pairs with $4|g_{uv}|$ larger than μ. The Berry curvature has modulus $\mu/2$ on anticommuting pairs whose product is in S and vanishes on all other pairs.

Proof. Proposition 3 with $|p| = \mu$ on S and $p = 0$ outside; for anticommuting u, v the product $P_u P_v$ is $\pm i P_{u+v}$, so $\langle P_u P_v \rangle$ is imaginary, of modulus $|p_{u+v}|$. ■

Proposition 9 (two mutually unbiased stabilizer states). Let s_1, s_2 be stabilizer state vectors whose stabilizer groups meet only in the identity, so that $|\langle s_1|s_2\rangle| = D^{-1/2}$, and let $\varphi = \theta + \arg\langle s_1|s_2\rangle$. The normalized state proportional to $s_1 + e^{i\theta} s_2$ is flat on a set that is not a stabilizer group exactly in the following cases: $N = 1$ with $\cos\varphi = (\sqrt{6} - \sqrt{2})/4$ or $-(\sqrt{6} + \sqrt{2})/4$ (all 3 observables, modulus $1/\sqrt{3}$); $N = 2$ with $\varphi = \pm\pi/2$ (12 observables, modulus $1/2$); $N = 3$ with $\varphi = \pm 3\pi/4$ (all 63, modulus $1/3$: the Hoggar fiducial is $\sqrt{8}$ times the uniform vector plus $(-2 + 2i)$ times the first basis vector); $N = 4$ with $\varphi = \pi$ (the 135 observables of a hyperbolic quadric, modulus $1/3$: $(-3, 1, \ldots, 1)$ is 4 times the uniform unit vector minus 4 times the first basis vector). There is none for $N \geq 5$.

Proof. Let $t = D^{-1/2}$ and $n = 2 + 2t\cos\varphi$ the squared norm. An observable of the first stabilizer group has expectation $\pm(1 + 2t\cos\varphi)/n$, and so has one of the second. Every other observable is, up to a phase, a product $P_1 P_2$ of non-identity elements of the two groups, and its expectation is $\pm 2t\cos\varphi/n$ if P_1, P_2 commute and $\pm 2t\sin\varphi/n$ if they anticommute. Equating the non-zero moduli: all three are equal only if $|\cos\varphi| = |\sin\varphi|$ and $1 - \sqrt{2}\,t = \sqrt{2}\,t$, so $D = 8$; with $\sin\varphi = 0$ one needs $1 - 2t = 2t$, so $D = 16$ and $\cos\varphi = -1$; with $\cos\varphi = 0$ one needs $1 = 2t$, so $D = 4$; for $D = 2$ there are no commuting products and $|1 + \sqrt{2}\cos\varphi| = \sqrt{2}|\sin\varphi|$ has the two stated solutions. The remaining case $1 + 2t\cos\varphi = 0$ gives a stabilizer state. ■

So the SIC fiducials of one and three qubits, a state with the Pauli spectrum of [LLY], and the quadric state are four members of one family, and that family stops at four qubits. The pivot

state is not in it: it is a combination of three pairwise unbiased stabilizer states.

5.3 Flat phase states, and a seven-qubit configuration

Proposition 10 (the support is a two-character set). Let ψ be flat on S and let $k^2 = |S|/(D-1)$, the inverse square of the modulus. For every point w,

$$\sum_{v\in S}(-1)^{\langle v,w\rangle} \;=\; D\,[w\in S] \;-\; k^2.$$

Equivalently, exactly $(D-2)|S|/(2(D-1)) + \frac{D}{2}[w\in S]$ observables of S commute with P_w. Hence the graph on $\mathbb{F}_2^{2N}$ in which u and v are adjacent when $u+v\in S$ is strongly regular (or complete), with eigenvalues $|S|$, $D-k^2$ and $-k^2$ [CK]. If $k^2 = D/2$ the sum takes the values $\pm D/2$: then S is a Hadamard difference set, the support of a bent function on the $2N$-dimensional space.

Proof. For a pure state ρ and any w, $\mathrm{tr}(\rho P_w \rho P_w) = \langle P_w\rangle^2$. Writing $\rho = D^{-1}\sum_v p_v P_v$ with $p_0 = 1$ and using $P_w P_v P_w = (-1)^{\langle v,w\rangle} P_v$ gives $p_w^2 = D^{-1}\sum_v (-1)^{\langle v,w\rangle} p_v^2$. Insert $p_v^2 = k^{-2}$ on S, 1 at $v=0$ and 0 elsewhere. ∎

The identity for p_w^2 holds for every pure state and is standard; we did not find its consequence for flat states stated. It accounts for the hyperplane numbers in Theorems 5 and 7. For the two four-qubit rows of Table 3 it gives a strongly regular graph with the parameters $(256, 135, 70, 72)$ of the affine polar graph of $Q^+(7,2)$; for the three-, five- and seven-qubit rows with modulus $2^{-(N-1)/2}$ it gives bent functions in 6, 10 and 14 variables.

Phase states. Let $a_j \in \mathbb{Z}_8$, $b_{jk}\in\mathbb{Z}_4$, $c_{jkl}\in\mathbb{Z}_2$ and

$$p(x) = \sum_j a_j x_j + 2\sum_{j<k} b_{jk}\,x_j x_k + 4\sum_{j<k<l} c_{jkl}\,x_j x_k x_l \pmod 8, \qquad |\Phi_p\rangle = D^{-1/2}\sum_x e^{i\pi p(x)/4}\,|x\rangle.$$

Φ_p is obtained from $|+\cdots+\rangle$ by T gates (a_j), controlled-S gates CS (b_{jk}) and CCZ gates (c_{jkl}); these are the diagonal gates of the third level of the Clifford hierarchy [CGK]. Let T be the symmetric trilinear form on $\mathbb{F}_2^N$ with $T(e_j,e_j,e_j) = a_j$, $T(e_j,e_j,e_k) = T(e_j,e_k,e_k) = b_{jk}$ and $T(e_j,e_k,e_l) = c_{jkl}$ modulo 2. Then $B(x,z) = T(x,x,z)$ is a symmetric bilinear form, alternating exactly when there is no T gate. A linear change of variables in T is a CNOT circuit on the state, and two polynomials with the same form differ by a diagonal Clifford gate.

Proposition 11 (phase states of the third level). For $a\neq 0$ let R_a be the radical and r_a the rank of the symmetric bilinear form $T(a,\cdot,\cdot)$. The expectation value of X^aZ^c in Φ_p has modulus $2^{-r_a/2}$ for the 2^{r_a} vectors c of one coset of $R_a^\perp$ and vanishes for every other c; diagonal observables have expectation value zero. So Φ_p has a single non-zero modulus exactly when $r_a = r$ is the same for every $a\neq 0$. Then the modulus is $2^{-r/2}$, the support has $(D-1)2^r$ points, r is even if $N\geq 2$, and Φ_p attains the floor on its support. If moreover $r = N-1$, each R_a has one non-zero element $\rho(a)$, the map ρ is an involution of the non-zero vectors, and

$$S = \{\,(a\mid c)\;:\; a\neq 0,\quad c\cdot\rho(a) = \epsilon(a)\,\}$$

for a function ϵ: the support of a bent function of Maiorana–McFarland type [McF].

Proof. Modulo 8 the third difference of p is $4T$, independent of the base point, and the second difference is even. So $p(x) - p(x+a) = h_a + 2q_a(x)$ with $q_a(0) = 0$ and $q_a(x+y) =$

$q_a(x) + q_a(y) + 2T(a,x,y)$ modulo 4, a $\mathbb{Z}_4$-valued quadratic form [CCKS], and $\langle X^a Z^c\rangle = D^{-1} e^{i\pi h_a/4} \sum_x i^{q_a(x)} (-1)^{c\cdot x}$. Now

$$\Big| \sum_x i^{q_a(x)} (-1)^{c\cdot x} \Big|^2 = \sum_z i^{q_a(z)} (-1)^{c\cdot z} \sum_y (-1)^{T(a,y,z)} = D \sum_{z\in R_a} i^{q_a(z)} (-1)^{c\cdot z}.$$

On R_a the map $z \mapsto i^{q_a(z)}$ is a character with values ± 1, so the last sum is 2^{N-r_a} when $(-1)^{c\cdot z}$ agrees with it on R_a and 0 otherwise. That r is even follows from Theorem 4. If $r = N-1$ then $T(a,\rho(a),\cdot) = 0$, which is symmetric in a and $\rho(a)$, so $\rho(\rho(a)) = a$. ■

Theorem 12 (all flat phase states of the third level on at most five qubits). Among the non-zero forms T in $N \le 5$ variables, those of constant rank are the following.

N	forms	of constant rank	classes under CNOT circuits	gates of a representative	r	observables
1	1	1	1	T	1	2
2	7	1	1	$CS(0,1)$	2	12
3	127	8	2	$CCZ(0,1,2)$; $CS(0,1)\ CS(0,2)$	2	28
4	16383	0				
5	33554431	166656	1	$CS(0,1)\ CS(1,3)\ CS(2,4)$ $CCZ(0,1,4)\ CCZ(0,2,3)$	4	496

For five qubits the stabilizer of the form in $GL(5,2)$ has order 60. Write Φ_5 for the state of the last row. The supports of the two three-qubit classes are equivalent under $Sp(6,2)$, and the support of Φ_5 is mapped onto the set S_5 of Theorem 7 by a symplectic map of $\mathbb{F}_2^{10}$. So no state of this kind is flat on four qubits; on five qubits every one of them is obtained from Φ_5 by a CNOT circuit and a diagonal Clifford gate; and the configuration found numerically in [Ci] is, up to a Clifford operation, the Pauli support of a circuit of five gates.

Proof. Exhaustive enumeration, with the orbits computed from the transvections that generate $GL(N,2)$ (`src/third_level.c`; for $N \le 4$ also `scripts/phase_states.py`). The two symplectic maps are found by search and verified on every point; the second is recorded in `results/phase_states.json`. ■

For $N = 2$ the state is that of [LLY] up to a phase and complex conjugation. In coordinates $X = (x_1, x_2, \alpha) \in \mathbb{F}_q \times \mathbb{F}_q \times \mathbb{F}_2$ with $q = 4$, the five-qubit form is

$$T(X,Y,Z) = \mathrm{Tr}\left[x_1y_1z_2^2 + x_1y_2^2z_1 + x_2^2y_1z_1 + x_2y_2z_1^2 + x_2y_1^2z_2 + x_1^2y_2z_2\right] + \alpha\,\sigma(y,z) + \beta\,\sigma(x,z) + \gamma\,\sigma(x,y),$$

where $\sigma(y,z) = \mathrm{Tr}[a\,(y_1z_2 + y_2z_1)]$, Tr is the trace to $\mathbb{F}_2$ and a generates $\mathbb{F}_4$. It is invariant under $(x_1, x_2) \mapsto (\lambda x_1, \lambda^{-1} x_2)$.

Theorem 13 (seven qubits). Let

$$|\Phi_7\rangle = CS_{01}\,CS_{23}\,CS_{45}\ \ CCZ_{012}\,CCZ_{035}\,CCZ_{124}\ \ CCZ_{046}\,CCZ_{056}\,CCZ_{136}\,CCZ_{236}\,CCZ_{256}\ |+\rangle^{\otimes 7}$$

and let S_7 be its Pauli support.

1. Φ_7 has expectation values $\pm 1/8$ on S_7 and 0 elsewhere, and $|S_7| = 8128 = 2^6(2^7 - 1)$.

2. S_7 contains 2 824 192 contexts and $d(S_7) = 729\,344$. In the optimal assignment every observable of S_7 lies on 504 more satisfied than violated contexts; the number of contexts through an observable is 1008, 1024, 1040, 1056, 1072 or 1104 (for 64, 3584, 1792, 1344, 896 and 448 observables).
3. Let $\mathcal{F}$ be the family of forms on $\mathbb{F}_2^7$ for which B is alternating of rank 6. The forms of constant rank in $\mathcal{F}$ are a single class under CNOT circuits, and the stabilizer of such a form in $GL(7,2)$ has order 7. In coordinates $\mathbb{F}_8 \times \mathbb{F}_8 \times \mathbb{F}_2$ the form is the one displayed above, with the trace of $\mathbb{F}_8$, with $a^3 + a^2 + 1 = 0$, and with the additional term $\mathrm{Det}(x_1, y_1, z_1)$, the alternating trilinear form of $\mathbb{F}_8$ over $\mathbb{F}_2$.
4. For each of the ten other types of B (alternating of rank 0, 2 or 4; not alternating, of rank 1 to 7) we tested 3×10^8 random forms; none has constant rank.

Proof. Item 1 and the counts of item 2 are finite computations in $\mathbb{Z}[i]$, made by three programs that share no code (Supplementary Section S2). The lower bound is Theorem 1 with $|S| = 8128$, $D = 128$: $\beta \le 126 \cdot 8128 \cdot 1016/(6 \cdot 127) = 1\,365\,504 = 2\,824\,192 - 2 \cdot 729\,344$. For item 3, choose a basis in which e_6 spans the radical of B and $e_0, \ldots, e_5$ is a symplectic basis of a complement. Every form of $\mathcal{F}$ is then the third difference of $2(x_0x_1 + x_2x_3 + x_4x_5) + 4w(x_0, \ldots, x_5) + 4x_6\, s(x_0, \ldots, x_5)$ for a set w of CCZ gates on the first six qubits (2^{20} choices) and a graph s on them, which must be nondegenerate (13888 choices). All 1.46×10^{10} such forms were tested (`src/family7.c`): 13 271 040 have constant rank. The group that preserves the normal form has order $64\,|Sp(6,2)| = 92\,897\,280$, and the solutions are one orbit of it, of size $92\,897\,280/7$ (`src/orbits7.c`). ∎

The noncontextuality inequality of S_7 has classical bound 1 365 504 and quantum value 2 824 192: no noncontextual model reproduces more than 74.2 per cent of its contexts. Item 4 is evidence, not proof, that Φ_7 is the only flat phase state of the third level on seven qubits. The numbers of observables for three, five and seven qubits, 28, 496 and 8128, are $2^{N-1}(2^N - 1)$; that they are the second, third and fourth perfect numbers reflects only that 7, 31 and 127 are prime. We do not know whether a flat phase state exists on nine qubits (it would have 130 816 observables of modulus 1/16): the forms with the symmetry $(x_1, x_2) \mapsto (\lambda x_1, \lambda^{-1} x_2)$ of the five- and seven-qubit ones do not have constant rank for nine or eleven qubits.

5.4 What does not attain the floor

For the $D^2 - D$ observables outside a maximal commuting set (the non-diagonal ones, say) there are $(D-1)(D-2)D^2/12$ contexts and Theorem 1 gives a violated fraction of at least $\frac{1}{2} - 2^{-N/2}$. By Theorem 4, equality would need a state with all non-diagonal expectation values of modulus $2^{-N/2}$, that is, a state whose orbit under the Pauli group forms, with the computational basis, a complete set of mutually unbiased bases. For two qubits this is the first row of Table 3. For four qubits and for every odd $N \ge 3$ no such state exists [K], so the degree is strictly above the floor there. A numerical search from random starts found flat four-qubit states only with $|S| = 135$ (both classes of Proposition 6) and flat five-qubit states only with $|S| = 496$; it found none with $|S| = 60$, or with $|S|$ equal to 124, 279 or 775. This is evidence, not proof, that Table 3 is complete up to five qubits.

6. Consequences for all N

Corollary A (full space). For every $N \ge 1$ every assignment of the $D^2 - 1$ Pauli observables violates at least $\varphi(D)\, L$ contexts (Lean: `full_space_fraction`). Since a uniformly random assignment violates $L/2$ contexts on average, $d/L \to 1/2$.

Exponential violation. The ratio of the quantum value to the noncontextual bound of the

full inequality satisfies $L/(L-2d) \geq (D+2)/(2\sqrt{D+1}) \sim 2^{N/2-1}$. Cabello [C] proposed this inequality in 2010 and argued for exponential growth; his Lemma 3 identifies the bound with the number of positive contexts, which is correct for two qubits but not for three, where 252 of the 315 predictions can be satisfied rather than 225 [M1]. Corollary A proves the growth.

Noise. If imperfections scale every correlator by a visibility η, the inequality is violated when $\eta > 1 - 2d/L$, hence whenever $\eta > 2\sqrt{D+1}/(D+2)$: above 0.6 for three qubits, 0.458 for four, 0.338 for five, 0.062 for ten. In the notation of [C, MS] the tolerated error per context $\varepsilon = 2d/L$ is at least $1 - 2\sqrt{D+1}/(D+2)$, which exceeds 0.707 (the largest value confirmed in the search of [MS], for configurations of a different kind) from six qubits on and tends to 1.

Proposition 4 (the floor is sharp up to a logarithmic factor). For every pure N-qubit state with $M = \max_{v \neq 0} |\langle P_v \rangle|$ there is an assignment of the full space with

$$\beta \;\geq\; \frac{(D-1)(D-2)}{6\,M^3}, \qquad \text{so} \qquad \frac{1}{(D+1)(D+2)\,M_N^3} \;\leq\; \frac{1}{2} - \frac{d}{L} \;\leq\; \frac{\sqrt{D+1}}{D+2},$$

where M_N is the least value of M over all states. $M_N \geq 1/\sqrt{D+1}$, with equality exactly when a Pauli-covariant SIC exists, and $M_N \leq C\sqrt{\log D / D}$ for an absolute constant C.

Proof. A pure state has $\operatorname{tr}\rho^3 = 1$; with $\rho = D^{-1}(I + \sum_v p_v P_v)$ and $\sum_{v \neq 0} p_v^2 = D - 1$, the trace computation of Theorem 1 gives $T(p) = \sum_\ell s_\ell\, p_u p_v p_w = (D-1)(D-2)/6$ (the cubic purity condition [JL, St]). Choose the signs $f(v)$ independently with mean p_v/M. Then $\mathbb{E}\,\beta(f) = T(p)/M^3$, and some f does at least as well. The upper bound on M_N follows from Lévy's lemma for Haar-random states and a union bound over the $D^2 - 1$ observables [HLW]. ■

So $\frac{1}{2} - d/L = \Theta(2^{-N/2})$ up to a factor $\log^{3/2} D$. In four sample runs the sign patterns of Haar-random states reached between 50 and 64 per cent of the floor value of β for four and five qubits. We did not find Proposition 4 in the notes we could read; we do not claim it is new.

Averaging between families. Every context of $W(7,2)$ lies on exactly 24 of its 120 elliptic quadrics, and every context of $Q^+(9,2)$ lies in exactly 112 of its 496 sections by an imaginary Pauli observable, each a signed copy of $W(7,2)$ [LGG, J1]; we re-verified these incidences by enumeration. With Theorem 2 and integer rounding this gives $d(Q^-(7,2)) \geq 291$, $d(W(7,2)) \geq 1455$, $d(Q^+(9,2)) \geq 6444$, $d(W(9,2)) \geq 28791$, each below the values of [J1, J4] and stated here only to show what the floor alone yields.

7. The data against the floor

Table 1 confronts every published or pre-printed value we know for the classical families with the floor. "Floor" is the integer bound of Theorem 1 (Theorem 2 for elliptic quadrics), computed in exact integer arithmetic. "Best assignment" is the smallest number of violated contexts known; all entries up to five qubits were reproduced by our own search (Supplementary Section S2).

configuration	N	points	contexts	averaging bound $\lceil L/5 \rceil$ [M2, LGG]	floor	best assignment	exact value
$W(3,2)$	2	15	15	3	2	3	3 [dB]
$Q^+(3,2)$	2	9	6	–	1	1	1
$W(5,2)$	3	63	315	63	**63**	63	63 [M1]
$Q^+(5,2)$	3	35	105	21	14	21	21 [dB]
$Q^-(5,2)$	3	27	45	9	**9**	9	9 [dB]
$W(7,2)$	4	255	5355	1071	1451	1575 [M2]	1575 [J1]*
$Q^+(7,2)$	4	135	1575	315	**315**	315 [M2]	315 [LGG]

configuration	N	points	contexts	averaging bound $\lceil L/5 \rceil$ [M2, LGG]	floor	best assignment	exact value
$Q^-(7,2)$	4	119	1071	215	291	315 [M2]	315 [J1]*
$W(9,2)$	5	1023	86955	17391	28786	31479 [M2]	open ($\geq$ 28830 [J4]*)
$Q^+(9,2)$	5	527	23715	4743	6426	6975 [M2]	6975 [J1]*
$Q^-(9,2)$	5	495	19635	3927	6501	7087 [M2]	open ($\geq$ 6510 [J4]*)
$W(11,2)$	6	4095	1396395	279279	527621	553140 [M2]	open
$Q^+(11,2)$	6	2079	365211	73043	120901	132391 [M2]	open
$Q^-(11,2)$	6	2015	332475	66495	125624	131700 [M2]	open
$W(13,2)$	7	16383	22362795	4472559	9227609	9406024 [M2]	open
$Q^+(13,2)$	7	8255	5720715	1144143	2161542	2331191 [M2]	open
$Q^-(13,2)$	7	8127	5458635	1091727	2252409	2294580 [M2]	open

* computer-assisted, not formally verified, not peer reviewed; we did not re-run these computations.

Reading of the table. (1) All 17 rows are consistent with the floor; it is attained in the three rows in bold and, after rounding, in $Q^+(3,2)$, the Mermin–Peres square. (2) From four qubits on (five for hyperbolic quadrics) the floor exceeds the averaging bound $L/5$, and the gap to the best assignment shrinks: the floor closes 75, 81, 91 and 96 per cent of the interval between $L/5$ and the best assignment for the full space at $N = 4, 5, 6, 7$. (3) For seven qubits the best known assignments of the full space and of the elliptic quadric are within two per cent of the floor. (4) Outside these families the floor is exact for the configurations of Table 3.

An empirical regularity, also recorded for four qubits in [J1]: in every best assignment we computed, $A + 3I$ has low rank. The spectra of A are $\{21 \pm 16\sqrt{6}, (-3)^{14}\}$ for $W(7,2)$, $\{(9 \pm 4\sqrt{23})^2, (-3)^{12}\}$ for $Q^-(7,2)$, $\{45 \pm 56\sqrt{2}, (-3)^{30}\}$ for $Q^+(9,2)$, and three (respectively three pairs of) eigenvalues different from -3 for $W(9,2)$ (respectively $Q^-(9,2)$).

Figure 1 shows the floor of Table 1 against the earlier bound and the best assignments; Figure 2 shows the spectra just described.

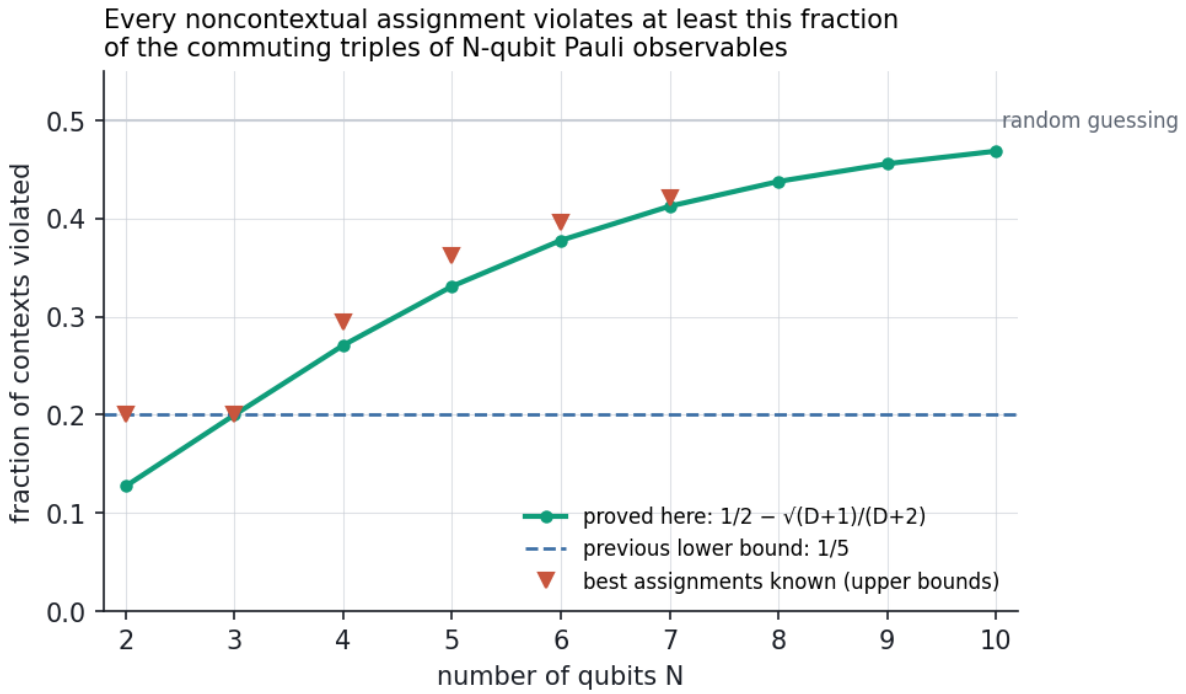


Figure 1. Lower bound on the fraction of violated contexts for the full N-qubit Pauli configuration, the earlier bound 1/5, and the best assignments known.

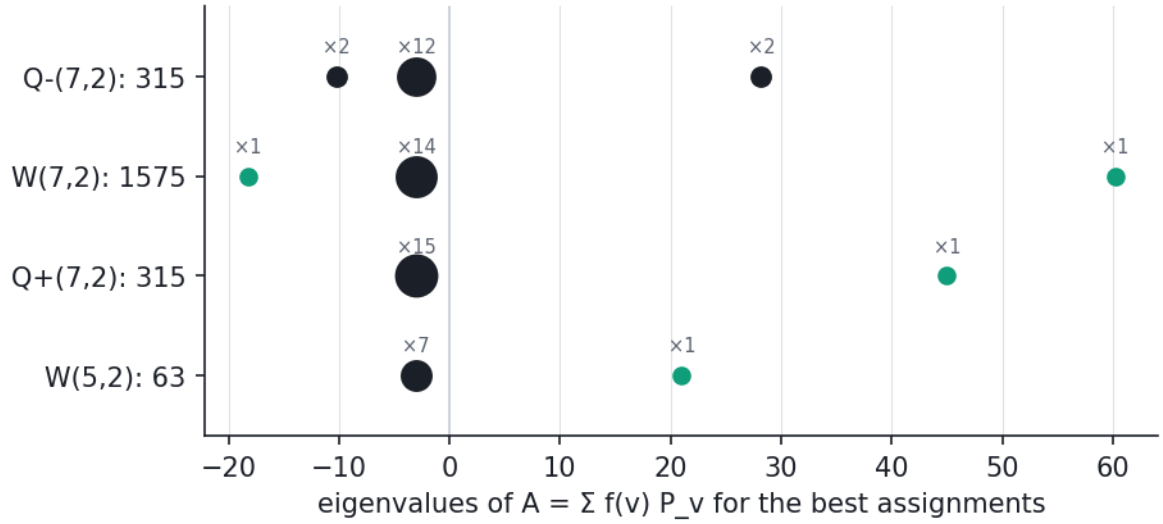


Figure 2. Spectra of $A = \sum f(v)P_v$ for the best assignments at three and four qubits.

8. The Lean 4 development

The sources are in `lean/HoggarBound/` (Lean 4.35.0, Mathlib v4.35.0).

module	content
`Moment.lean`	`sum_cube_le` (the scalar inequality); `trace_pow_eq_sum_eigenvalues_pow` ($\operatorname{tr} A^k = \sum \lambda_i^k$ for Hermitian A); `trace_cube_le`

module	content
`Pauli.lean`	bit vectors `Fin N → ZMod 2`; Weyl operators and `pauli u` as explicit complex matrices; `pauli_mul`, `context_product` ($P_u P_v P_{u+v} = \tau(u,v) I$), `pauli_comm`; traces of one, two and three Pauli operators; $\tau = \pm 1$ on commuting pairs
`Bound.lean`	`flags` (ordered commuting pairs with product in S: six per context), `beta6`, the operator `opA`; `trace_opA_cube` ($\operatorname{tr} A^3 = D\,\beta_6$); `hoggar_bound`
`Counting.lean`	`violated`, `beta6_eq_card`; the flag count of the full space for every N (`card_flags_all`); `violatedLines` and `card_violated_le`; `four_qubit_contexts`
`Corollaries.lean`	`full_space_bound`, `full_space_fraction` (all N); `three_qubit_contexts`, `five_qubit_contexts`
`Quadric.lean`	the real quadric; its point and flag counts for every N by character sums; `four_qubit_hyperbolic_contexts`, `five_qubit_hyperbolic_contexts`
`MaskConfig.lean`	configurations and assignments given by bit masks; `card_flags_cfg`, `card_violated_cfg` (their flag counts as sums the kernel can evaluate); `degree_of_certificate`
`FlatData.lean`, `SaturatedFive.lean`	generated: the masks of the first six rows of Table 3, and the five-qubit counts in 32 blocks
`Saturated.lean`	`pivot_degree`, `cieslinski5_degree`, `quadric_degree`, `hoggar_degree`, `ccz_degree`, `lly_degree`
`PhaseData.lean`, `SaturatedPhaseFive.lean`	generated: the masks of Φ_5 and Φ_7; `phase5_degree`

The main theorem, as checked:

```
theorem hoggar_bound (hN : 1 ≤ N) (S : Finset (Pt N)) (h0 : (0 : Pt N) ∉ S) (f : Pt N → ℝ)
    (hf : ∀ u ∈ S, f u = 1 ∨ f u = -1) :
    ((2 : ℝ) ^ N - 1) * beta6 S f
      ≤ ((2 : ℝ) ^ N - 2) * S.card * Real.sqrt (((2 : ℝ) ^ N - 1) * S.card)
```

Here `beta6 S f` is six times $\beta_S(f)$, a sum over ordered pairs, and the sign of a context is by definition the real part of the scalar $\tau(u,v)$ with `pauli u * pauli v * pauli (u + v) = tau u v • 1`. Table 2 lists the corollaries.

Lean theorem	statement
`full_space_fraction`	for all $N \geq 1$: #violated flags $\geq \left(\frac{1}{2} - \sqrt{2^N+1}/(2^N+2)\right)$ #flags
`three_qubit_contexts`	every assignment of the 63 observables violates at least 63 contexts
`four_qubit_contexts`	every assignment of the 255 observables violates at least 1451 contexts
`five_qubit_contexts`	every assignment of the 1023 observables violates at least 28786 contexts
`four_qubit_hyperbolic_contexts`	every assignment of the 135 real observables violates at least 315 contexts
`five_qubit_hyperbolic_contexts`	every assignment of the 527 real observables violates at least 6426 contexts
`pivot_degree`	for the 135 observables of S_B: every assignment violates at least 75 contexts, and some assignment violates exactly 75

Lean theorem	statement
`cieslinski5_degree`	for the 496 observables of S_5: at least 480, and exactly 480 for some assignment
`phase5_degree`	the same statement for the 496 observables in the support of the circuit state Φ_5
`quadric_degree`, `hoggar_degree`	the same two-sided statements with 315 (four-qubit hyperbolic quadric) and 63 (three qubits)
`ccz_degree`, `lly_degree`	some assignment of the 28 (respectively 12) observables violates no context

`#print axioms` reports `[propext, Classical.choice, Quot.sound]` for each of them: no `sorry`, no `native_decide`, no additional axiom.

How the exact degrees are checked. The statement of `pivot_degree` is about a set of 135 explicit complex 16×16 matrices, given by a 256-bit mask: bit $a + 16c$ says whether $P_{(a|c)}$ belongs to it. `MaskConfig.lean` proves, for every N and every mask, that the number of flags of the configuration and the number of flags violated by an assignment that is itself given by a mask are double sums over codes of Boolean expressions built from bit tests, exclusive or and addition of natural numbers. The kernel evaluates these sums (`decide +kernel`); for five qubits each sum has 2^{20} terms and is evaluated in 16 blocks. `degree_of_certificate` then combines three evaluated numbers (observables, flags, flags violated by the witness) with `hoggar_bound`. That the masks are the supports and sign patterns of the states of Table 3 is checked by the programs of Supplementary Section S2 in exact arithmetic; it is not needed for, and not part of, the Lean statements. For seven qubits the two sums have 2^{28} terms each and were not evaluated in the kernel: the degree 729 344 of S_7 rests on `hoggar_bound`, which needs only the number 8128, and on a count of contexts and violated contexts made by programs.

What is not formalized. Theorem 2 (the quaternionic form) and all elliptic-quadric numbers; Theorems 3 and 4 as equivalences, and every statement about states, spectra or geometry (the Lean results are the inequality and the two-sided degree statements); Propositions 4, 6 and 8 to 11; the enumerations of Theorems 12 and 13 and the seven-qubit counts; the limit statement as a theorem about limits (the inequality for each N is formal); the exact values of [J1]. For the classical families beyond three qubits and $Q^+(7, 2)$, upper bounds are witnesses checked by the programs of Supplementary Section S2.

Independent audit. A separate set of agent sessions, working concurrently in the same repository and without editing our files, re-checked the first six modules from source, confirmed the axiom report, and added two modules of its own (`IndependentContexts.lean`, `IndependentWitness.lean`). They prove that a violated context carries exactly six violated flags (used in `MaskConfig.lean`) and, by kernel evaluation of an integer certificate, that the explicit assignment "$f(a, c) = +1$ if $a = 0$ or $c = 0$ or the number of Y factors is at least 2, and -1 otherwise" violates exactly 63 contexts. Their reports are in `results/lean-audit.md`, `results/literature-audit.md` and `results/stack-independent-audit.md`, and their companion note is `paper/independent-research-note.md`. The literature audit is what first pointed us to [J1]. The modules from `MaskConfig.lean` on, and everything in Section 5, were written after that audit and are not covered by it.

9. Limitations and open problems

- Novelty is claimed to the extent of a bounded search (arXiv and Zenodo queries on 9 November 2026, and the sources of Section 1.2). The field moved quickly in October 2026, and several of the relevant preprints are less than a month old. Statements of the

form "to our knowledge" mean exactly that.

- The configurations S_B, S_5 and S_7 are defined by states, not by a classical geometric construction. We do not know an infinite family of non-closed configurations that attain the floor: Proposition 9 shows that the simplest family of states stops at four qubits, and for flat phase states we have three, five and seven qubits only, with nine open. Theorem 12 is complete for its class of states; whether Table 3 is complete for all states on $N \leq 5$ qubits is supported by numerical search only. Whether the states of the odd class of Proposition 6 form a single Clifford orbit was not determined.
- For seven qubits the uniqueness statement is proved only inside the family $\mathcal{F}$ of Theorem 13; outside it there is a random census. The enumeration behind it (and the one of Theorem 12) is a computation, repeated here by one program only, apart from the agreement of the count with the orbit size and the second implementation for $N \leq 4$. The seven-qubit counts of contexts are checked by three programs but not by the Lean kernel.
- Within the classical families the floor is not attained beyond the three listed cases [J2]. Exact values for five or more qubits are open except $Q^+(9, 2)$ [J1]; the floor alone leaves the intervals of Table 1.
- The exact four- and five-qubit values of the classical families rest on computer-assisted work that is neither formally verified nor peer reviewed [J1]. A formal proof of $d(W(7, 2)) = 1575$ would be a natural next target; our Lean development gives the lower bound 1451 only.
- The quaternionic form (Theorem 2) and the equivalences of Theorems 3 and 4 are proved on paper and tested numerically, not formalized.
- Nothing here is an experiment on quantum hardware, and nothing shows a computational advantage. The simulation demonstrates what quantum mechanics predicts and what no noncontextual model can do.

10. How the work was produced

The research direction, derivations, programs, Lean proofs and text were produced on 9 November 2026 by an AI system, Claude Fable 5 (Anthropic), used through the Claude Code agent, at the request and under the direction of the author. Within the author's brief the system chose the problem, searched the literature, derived the statements, wrote and ran the programs and the Lean proofs, and wrote the text. Its output was validated as follows: the theorems of Section 8 by the Lean kernel; every number in the paper by a program that counts in exact integer arithmetic, and the main ones by two or three programs that share no code; every reference against Crossref, arXiv, Zenodo or the publisher's site; and the first part of the work by a second group of AI agent sessions working independently (Section 8).

Data availability statement

All data, programs and formal proofs that support the findings are openly available at https://github.com/0thernes/hoggar-bound-lab (tagged releases; the repository is archived by Software Heritage under that address).

Acknowledgments

AI disclosure statement. Claude Fable 5 (Anthropic) was used for the research itself and for the preparation of the manuscript, as described in the previous paragraph. No human has refereed the mathematics. The author directed the work and is responsible for its release; the Lean statements are checked by the Lean kernel, and everything else should be read as an

unrefereed preprint.

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Licence of the supporting material. The repository is distributed under a verification licence (see its `LICENSE` file). Third-party components (Moonlab, quantum_rng, quantum_geometric_tensor) are MIT-licensed by their authors and are vendored with their licences and provenance records.

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